Student Lifestyle and Academic Productivity

Associations between sleep, physical activity, recreational screen time, study habits and well-being among university students

Course project report | Introduction to Data Handling and Visualization | October 2026

Group members: [add names and roll numbers here]

Abstract

This study examines how sleep, physical activity, recreational screen time and study habits relate to academic productivity, stress and concentration among university students. An anonymous 21-question Google Form was shared through student groups between 18 and 21 October 2026 and produced 59 submissions, of which 54 were valid. The raw data contained blank submissions, free-text numeric answers and impossible values, which were handled with a documented pipeline: range rules, text parsing, k-nearest-neighbour imputation of predictors only, and z-score, min-max and log normalization. Descriptive statistics, Spearman correlations with bootstrap confidence intervals, Kruskal-Wallis tests and a standardized regression were used, with Benjamini-Hochberg correction for multiple testing. Concentration was the outcome most clearly linked to habits: it rose with study consistency (rho = 0.45) and sleep quality (rho = 0.41). Sleep duration, physical activity and screen time showed no significant association with stress, productivity or CGPA, although students with more recreational screen time reported more phone distraction (grouped comparison adjusted p = 0.018; correlation rho = 0.34, not significant after correction). The findings are limited by a small, male-dominated convenience sample and self-reported measures.

1. Introduction

1.1 Background and motivation

University life asks students to balance coursework with sleep, exercise and heavily digital leisure. Late-night phone use, long recreational screen time and irregular study routines are widely blamed for poor focus and wellbeing, while exercise and good sleep are widely recommended. In practice, however, students rarely know which of these habits matter most for how focused and productive they actually feel. Evidence drawn from their own campus is more persuasive than general advice, and a small, carefully designed survey is a practical way to obtain it. This project follows the full data handling and visualization lifecycle, from research design and data collection through cleaning, exploratory analysis and communication of findings.

1.2 Research question

How are sleep, physical activity, recreational screen time and study habits associated with academic productivity, stress and concentration among university students?

1.3 Hypotheses

- **H1:** students who exercise more report lower stress and higher concentration and productivity.
- **H2:** better and longer sleep is associated with higher concentration and productivity.
- **H3:** higher recreational screen time is associated with lower productivity and more phone distraction.
- **H4:** more consistent and longer study routines are associated with higher CGPA and productivity.

The report is organised as follows: Section 2 describes the study design and data collection, Section 3 the preprocessing pipeline, Section 4 the descriptive and exploratory results with visualizations, and Section 5 the conclusions and limitations. The final questionnaire is in Appendix A.

2. Methodology

2.1 Research design

The study is a cross-sectional, quantitative survey with both numeric and categorical items. Lifestyle habits (physical activity, sleep, digital behavior, study habits) are treated as explanatory variables, and perceived stress, concentration, productivity and CGPA as outcomes. All measures are self-reported, which suits the aim of capturing students' own experience but introduces recall and reporting bias (Section 5.3).

2.2 Questionnaire design

The questionnaire has 21 questions in six blocks: profile, physical activity, sleep and rest, digital behavior, study habits and academics, and well-being. Design choices follow standard survey practice: one idea per question (social media and gaming are asked separately), banded options for gym and sports frequency because exact recall is unreliable, labeled 1 to 5 rating scales for subjective states, neutral wording, and outcome questions placed last. The full instrument is in Appendix A.

Table 1. Variables collected

Block	Variables	Type and coding
Profile	Year of study, gender	Categorical (year merged into three groups for analysis)
Physical activity	Gym days per week, sports hours per week	Ordinal bands, coded to band midpoints
Sleep and rest	Sleep hours, bedtime, sleep quality, phone use before sleep	Numeric hours; ordinal bands; 1-5 rating; ordinal (Never/Sometimes/Always)
Digital behavior	Recreational screen, social media and gaming/OTT hours; phone distraction	Numeric hours (free text); 1-5 rating
Study and academics	Study hours, study consistency, attendance, CGPA	Numeric; 1-5 rating; ordinal bands; numeric (0-10)
Well-being	Stress, concentration, productivity, tiredness; perceived benefit of activity	1-5 ratings; categorical (4 options)

2.3 Sampling and data collection

The target population is university students. The form was shared through class and hostel group chats and completed voluntarily between 18 and 21 October 2026, giving a **convenience (voluntary response) sample**. This method was chosen because no sampling frame of students was available and it allowed the group to reach more than 50 respondents within the project timeline at no cost. Its drawback is that respondents are self-selected and not representative, so results describe this sample rather than students in general. A sample of 54 gives adequate power only for moderate or stronger associations: at 80% power and alpha = 0.05, a Spearman correlation of about 0.37 is the smallest reliably detectable effect.

2.4 Ethical considerations

Participation was voluntary and anonymous. No names or email addresses were collected, the introduction explained the purpose of the study, and responses are used only for this coursework.

3. Data Preprocessing

3.1 Data organization

The Google Form export (*raw_data.csv*, 59 rows and 22 columns) was kept unmodified. All cleaning is scripted in *preprocessing.py* so every step is reproducible; the output is *cleaned_data.csv* (54 rows). Long question texts were replaced with short variable names, and a coding scheme was defined for each ordinal variable (Section 3.6).

3.2 Removing unusable submissions

Four submissions contained only a timestamp, and one contained profile answers but none of the outcome questions (stress, concentration, productivity). Outcome variables cannot be imputed without inventing the result, so these five rows were removed, leaving **54** valid responses. No exact duplicate responses were found.

3.3 Standardizing categories and parsing free-text numbers

Gender labels differed only in case (male and Male) and were standardized. Because the form did not enforce numeric input, the hour questions arrived as free text such as '5-6 hours', '6 to 7 hrs', '30 minutes' or 'No time'. A parser converted ranges to their midpoint (5-6 hours becomes 5.5), minutes to hours (30 minutes becomes 0.5) and 'No time' to 0. Answers with no usable number ('na', '1/3') were treated as missing. Year of study was merged into three groups (1st-2nd, 3rd, 4th-year or postgraduate) because some levels had only one to three students.

3.4 Outliers and impossible values

Plausibility ranges were set for each numeric variable: sleep 2-14 h, screen, social media and gaming 0-16 h, study 0-14 h and CGPA 4-10. Six entries fell outside these ranges and were set to missing rather than deleted, so the rest of each student's answers were retained: 99 h of screen time, 100 h of social media, 1000 h of gaming, 100 h of study (entered as -100), 0 h of sleep and a CGPA of 1000. Plausible but extreme values (for example 16 h of screen time) were kept because they are within what the question allows.

3.5 Missing values

Table 2. Missing values remaining after cleaning (n = 54)

Variable	Missing	Main cause
Sleep hours	6	Left blank (5), impossible value 0 h (1)
CGPA	6	Left blank, 'na', or value 1000
Study hours	3	Left blank or -100
Gaming/OTT hours	2	Ambiguous '1/3' and value 1000
Screen time	1	Value 99
Social media hours	1	Value 100
All rating and categorical questions	0	None

After cleaning, 19 of 324 cells (5.9%) in the six numeric columns were missing; 11 students had at least one gap and only 3 had two or more, so there is no sign of a systematic pattern. The strategy was: (i) **outcomes are never imputed** (stress, concentration, productivity, CGPA); each analysis uses students who answered both variables (pairwise deletion, n = 47 to 54); (ii) missing **predictor** hours were imputed with k-nearest-neighbours (k = 5, distance-weighted) using related lifestyle variables, because neighbours with similar sleep, activity and study patterns give more realistic values than one global median; (iii) every imputed cell is flagged. As a **sensitivity check**, the 18 planned correlations were recomputed on the imputed data: the largest change in any correlation was 0.06 and no conclusion changed significance (0 of 18).

3.6 Encoding, reverse coding and derived variables

Banded answers were converted to numbers: gym days (0, 1.5, 3.5, 5.5 per week) and sports hours (0, 2, 5, 8 per week) by band midpoint, and bedtime, phone use before sleep and attendance by ordered codes. Stress, tiredness and distraction were reverse-coded (6 minus score) where composites were built so that higher always means better. Derived variables include total physical activity (gym plus sports), an activity level (Inactive: no gym or sports; Moderate: up to 4 combined units; Active: more), groups for sleep (under 6 h, 6-8 h, over 8 h) and screen time (up to 3 h, 3-6 h, over 6 h), and an academic score (mean of productivity and concentration, scaled 0-100). A wellbeing composite of stress, tiredness and sleep quality was tested but its reliability was poor (Cronbach's alpha = 0.36), so those items are analyzed separately.

3.7 Normalization and transformation

Three versions of the continuous variables were created: **z-scores** (mean 0, SD 1), used in the regression so coefficients are comparable; **min-max scaling** to a 0-1 range, for comparing variables on one scale; and a **log(1 + x) transformation** for the three right-skewed hour variables. The log transformation reduced skewness for sleep from 1.49 to 1.03, social media from 1.09 to -0.49 and gaming from 1.26 to 0.09. Because several variables remain skewed and the ratings are ordinal, rank-based (Spearman and Kruskal-Wallis) methods are used for inference.

3.8 Consistency checks

Logical checks were added as flag columns rather than deleting rows: 3 students reported more social media time than total screen time, 2 reported sleep, study and screen time summing to over 24 hours a day, 2 said they do no physical activity yet reported some gym or sports, and 1 gave the same answer to every rating item. Flagged rows were retained because the analyses are rank-based and robust to a few such cases.

4. Results and Analysis

4.1 Descriptive statistics

Table 3. Descriptive statistics of numeric variables (cleaned data, no imputation)

Variable	n	Mean	Median	SD	Min-Max	Skew
Sleep (h/night)	48	6.81	6.25	1.53	5.0 - 12.0	1.49
Recreational screen time (h/day)	53	6.50	6.00	3.65	0.0 - 16.0	0.65
Social media (h/day)	53	3.30	3.00	2.00	0.2 - 10.0	1.09
Gaming / OTT (h/day)	52	1.65	1.25	1.62	0.0 - 7.0	1.26
Study time (h/day)	51	2.74	2.50	1.62	0.0 - 7.0	0.87
CGPA (out of 10)	48	8.36	8.45	0.77	6.0 - 9.6	-0.80

Table 4. Self-rated items (1 = lowest, 5 = highest)

Rating (1-5)	Mean	Median	% rating 4-5	% rating 1-2
Sleep quality	3.13	3	39%	26%
Stress	3.33	3	46%	20%
Tiredness	3.48	4	50%	15%
Phone distraction	3.48	4	52%	17%
Study consistency	2.54	3	17%	44%
Concentration	2.93	3	26%	26%
Productivity	2.85	3	22%	33%

Sample profile (Figure 1). Of 54 students, 32 (59%) are in third year, 16 in second year, 3 in first year, 2 in fourth year and 1 are postgraduates. 48 (89%) are male, 5 female and 1 preferred not to say, so gender comparisons are not meaningful. Physical activity is evenly split: 17 inactive, 19 moderate and 18 active students; 44% never go to the gym and 50% play no sports. Late nights and phone habits are common: 59% fall asleep after 1 AM, and 80% always use their phone before sleeping.

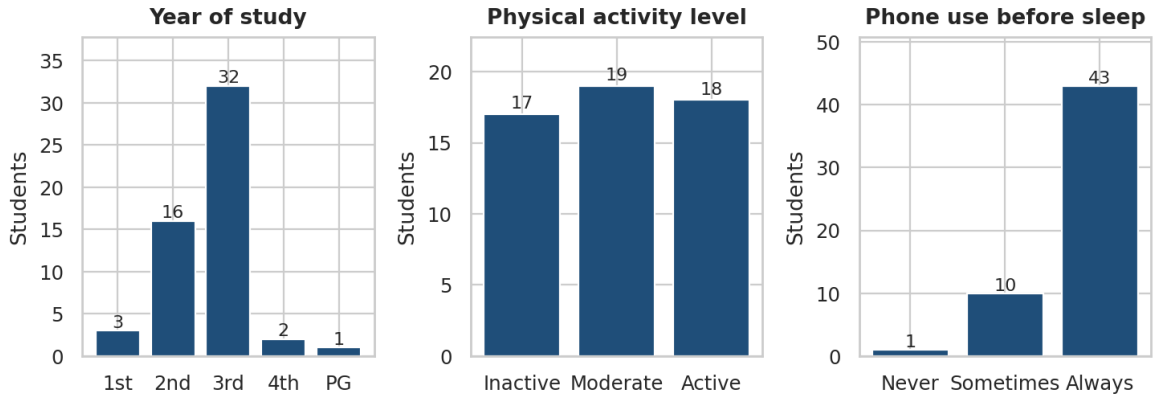


Figure 1. Sample profile: year of study, physical activity level (Inactive: no gym or sports; Moderate: up to 4 combined units; Active: more) and phone use in the 30 minutes before sleep.

Distributions (Figure 2). Median sleep is 6.25 hours, and 60% of the students who answered sleep under 7 hours. Median recreational screen time is 6 hours a day and 47% exceed 6 hours; social media accounts for a median of 3 hours. Students study a median of 2.5 hours a day, and CGPA is high and left-skewed (median 8.45). Hour variables are right-skewed, which supports the use of medians and rank-based methods. Self-rated well-being is strained: 46% rate their stress 4 or 5, 50% their tiredness and 52% their phone distraction, while only 17% rate their study routine as highly consistent (44% rate it 1 or 2).

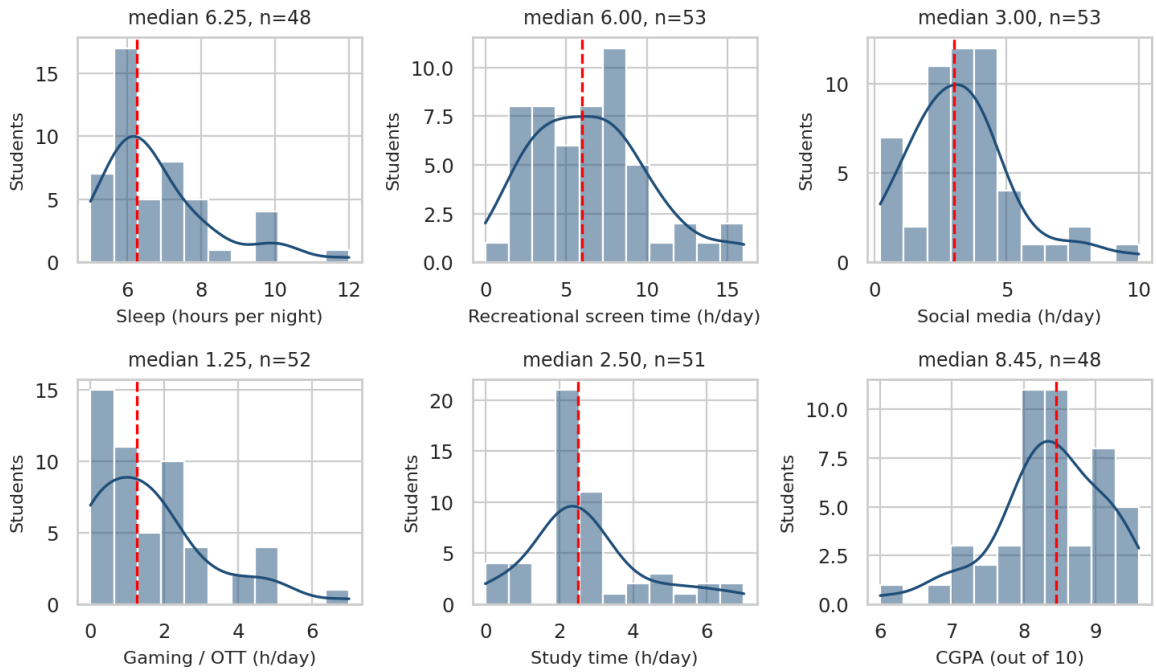


Figure 2. Distributions of the six numeric variables. The dashed red line marks the median.

4.2 Relationships between variables

Figure 3 shows all pairwise Spearman correlations. The strongest links are between lifestyle measures themselves: screen time and social media time (rho = 0.55), and sleep hours and sleep quality (rho = 0.46). Among outcomes, concentration is the variable most connected to habits: it rises with study consistency (rho = 0.45, 95% CI 0.17 to 0.67) and sleep quality (rho = 0.41) and falls with phone distraction (rho = -0.30). Stress and tiredness move together (rho = 0.38), and concentration and productivity are positively related. CGPA is not meaningfully correlated with any lifestyle variable. Because 91 pairs were examined, p-values were adjusted with the Benjamini-Hochberg false-discovery procedure (Table 5); only three pairs remain significant after adjustment: screen time with social media, sleep hours with sleep quality, and study consistency with concentration.

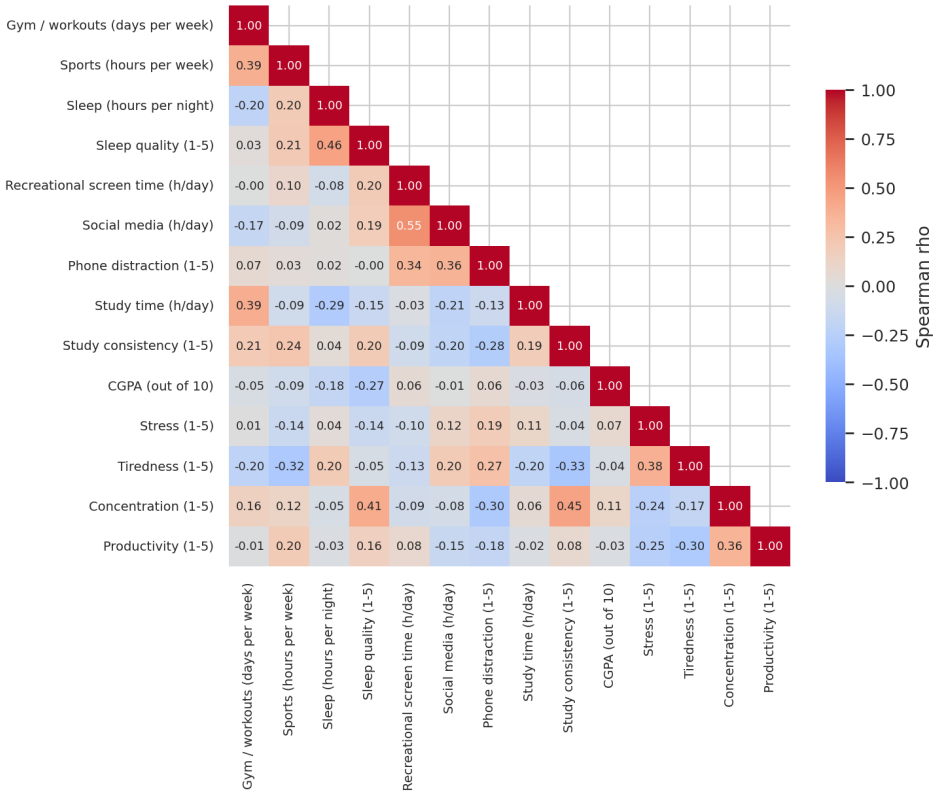


Figure 3. Spearman correlation matrix (pairwise complete observations). Red is positive, blue is negative.

Table 5. The eight strongest correlations among all 91 pairs, with false-discovery-adjusted p-values

Pair (Spearman)	n	rho	p	Adjusted p
Screen time and social media time	53	0.55	<0.001	0.002
Study consistency and concentration	54	0.45	<0.001	0.028
Sleep hours and sleep quality	48	0.46	0.001	0.035
Sleep quality and concentration	54	0.41	0.002	0.053
Gym days/week and sports h/week	54	0.39	0.004	0.065
Gym days/week and study hours	51	0.39	0.004	0.065
Stress and tiredness	54	0.38	0.005	0.065
Social media time and phone distraction	53	0.36	0.008	0.080

4.3 Physical activity (H1)

Students in the Active group report higher concentration (mean 3.17) than Inactive (2.82) and Moderate (2.79) students, and lower stress (3.17 versus 3.53 for Inactive), but the confidence intervals overlap widely (Figure 4) and Kruskal-Wallis tests find no significant difference for stress (p = 0.79), concentration (p = 0.42), productivity (p = 0.10) or CGPA (p = 0.50). The Moderate group has the lowest productivity (2.42), but this is not significant (p = 0.098, adjusted p = 0.80) and does not follow a dose-response pattern. Total activity is only weakly related to concentration (rho = 0.19, p = 0.16). Interestingly, students who go to the gym more also study more (rho = 0.39, unadjusted p = 0.004; adjusted p = 0.06, exploratory). Perception differs from the data: 59% of students say activity helps their studies at least somewhat, although 28% do none.

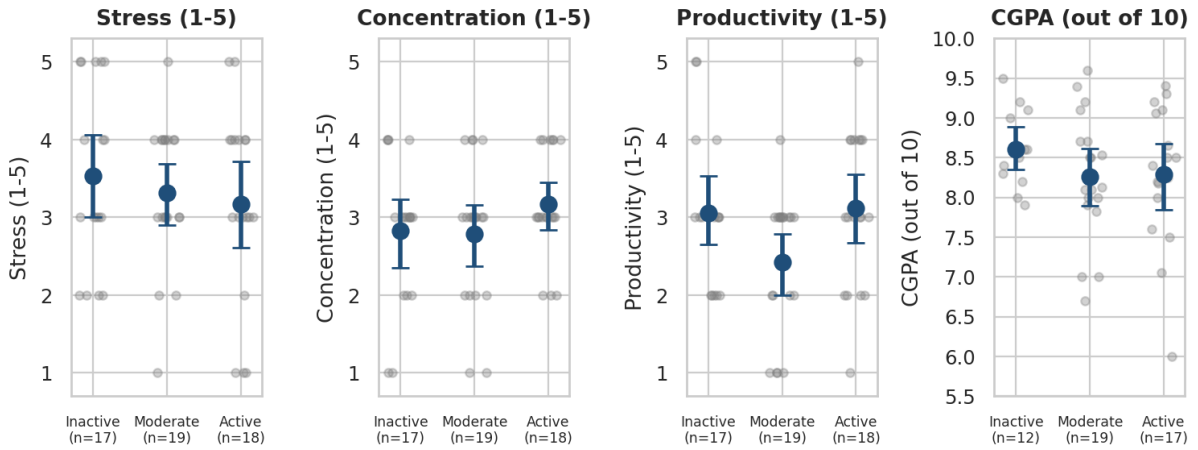


Figure 4. Outcomes by physical activity level. Dots are individual students; the large marker is the group mean with a 95% bootstrap confidence interval. No difference is statistically significant.

4.4 Sleep and study habits (H2 and H4)

Sleep *duration* shows no association with concentration (rho = -0.05), stress or productivity (all |rho| below 0.05), whereas sleep *quality* does: mean concentration rises from 2.17 among students rating their sleep 1 to 3.83 among those rating it 5 (rho = 0.41, 95% CI 0.15 to 0.63, adjusted p = 0.042; Figure 5). Study consistency shows an even clearer gradient, from 2.46 (rating 2) to 3.67 (rating 4-5; rho = 0.45, adjusted p = 0.028 across all 91 pairs). Neither consistency nor study hours relates to CGPA (Figure 6; rho = -0.06 and -0.03), so concentration, not grade, is where habits show up in this sample.

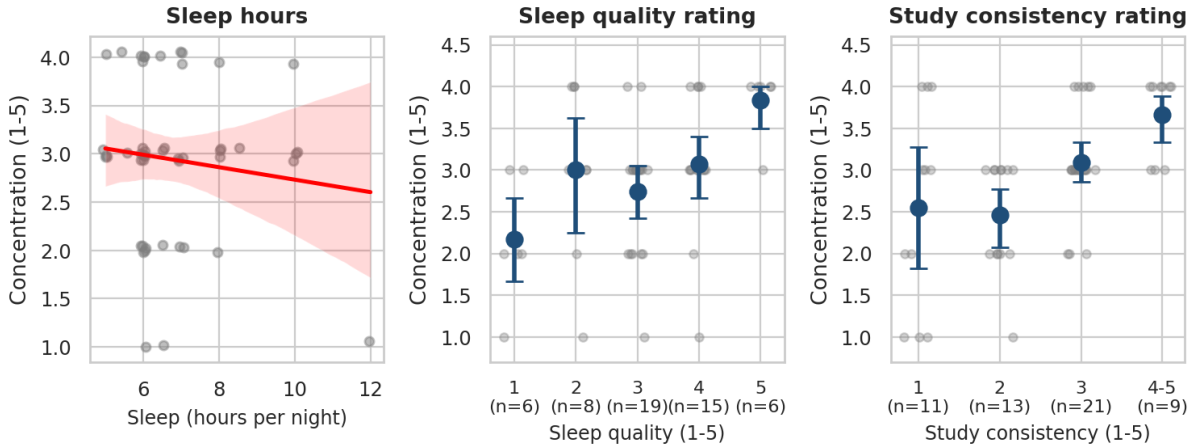


Figure 5. Concentration against sleep hours (scatter with linear fit and 95% band), sleep quality rating and study consistency rating (means with 95% bootstrap intervals; consistency levels 4 and 5 are combined because only one student chose 5).

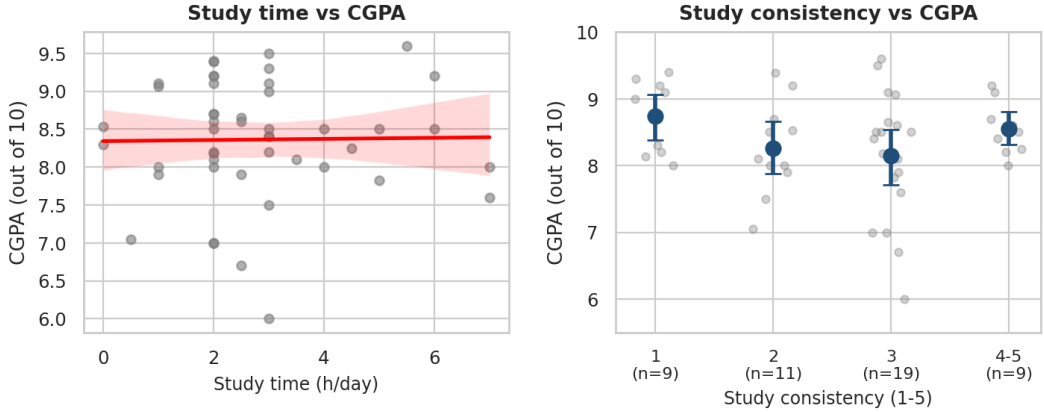


Figure 6. CGPA against daily study time and study consistency. Neither shows a meaningful relationship.

4.5 Recreational screen time (H3)

Screen time is related to *distraction* but not to the outcomes. Students with up to 3 hours of recreational use (n = 11) report the least phone distraction (2.55) compared with 3.94 for 3-6 hours (n = 17) and 3.52 for over 6 hours (n = 25). This group difference is statistically significant (Kruskal-Wallis H = 13.2, p = 0.001, adjusted p = 0.018 across the 13 group tests, epsilon-squared = 0.22), although it was added after the planned tests and the low-screen group is small. The continuous relationship is weaker: screen hours correlate with distraction at rho = 0.34 (p = 0.013), which does not survive correction across the 18 planned correlations (adjusted p = 0.12; Figure 7). Screen time shows no association with productivity (rho = 0.08) or concentration (rho = -0.09); the medium-screen group has the lowest productivity (2.41), but group differences in productivity are not significant (p = 0.13).

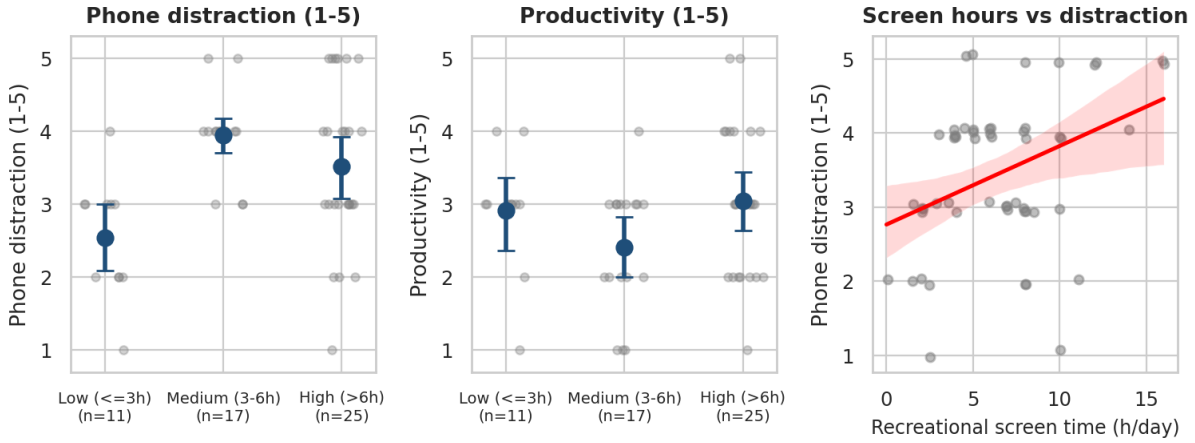


Figure 7. Phone distraction and productivity by recreational screen-time group (means with 95% bootstrap intervals), and screen hours against distraction with a linear fit.

4.6 Summary of hypothesis tests

Table 6 lists the 15 planned tests tied to the hypotheses (from a family of 18 planned correlations; adjusted p-values use Benjamini-Hochberg over all 18) plus the exploratory consistency-concentration pair found in the full scan. Confidence intervals are 95% bootstrap intervals.

Table 6. Spearman correlation tests by hypothesis

Hypothesis	Pair	n	rho	95% CI	p	Adj. p	Verdict
H1	Physical activity (gym + sports) - Stress	54	-0.12	-0.41 to 0.20	0.401	0.837	Not supported
H1	Physical activity (gym + sports) - Concentration	54	0.19	-0.09 to 0.45	0.163	0.731	Not supported
H1	Physical activity (gym + sports) - Productivity	54	0.12	-0.18 to 0.42	0.391	0.837	Not supported
H1	Physical activity (gym + sports) - CGPA	48	-0.08	-0.35 to 0.19	0.572	0.837	Not supported
H2	Sleep hours - Stress	48	0.04	-0.26 to 0.30	0.807	0.837	Not supported
H2	Sleep hours - Concentration	48	-0.05	-0.32 to 0.23	0.731	0.837	Not supported
H2	Sleep hours - Productivity	48	-0.03	-0.32 to 0.28	0.837	0.837	Not supported
H2	Sleep quality - Concentration	54	0.41	0.15 to 0.63	0.002	0.042	Supported
H3	Screen time - Productivity	53	0.08	-0.19 to 0.36	0.555	0.837	Not supported
H3	Screen time - Concentration	53	-0.09	-0.37 to 0.21	0.509	0.837	Not supported
H3	Screen time - Phone distraction	53	0.34	0.02 to 0.60	0.013	0.118	Suggestive only
H3	Social media time - Concentration	53	-0.08	-0.35 to 0.22	0.587	0.837	Not supported
H4	Study hours - CGPA	47	-0.03	-0.31 to 0.25	0.827	0.837	Not supported
H4	Study consistency - CGPA	48	-0.06	-0.33 to 0.21	0.680	0.837	Not supported
H4	Study consistency - Productivity	54	0.08	-0.24 to 0.40	0.580	0.837	Not supported
Exploratory	Study consistency - Concentration	54	0.45	0.17 to 0.67	<0.001	0.028*	Supported

** adjusted across all 91 pairs in the exploratory scan; other adjusted p-values are across the 18 planned tests. The grouped comparison of distraction by screen-time group is reported separately in Section 4.5.*

4.7 Multiple regression

To see which habits matter most when considered together, the academic score (mean of productivity and concentration, scaled 0-100) was regressed on the five standardized predictors with heteroscedasticity-robust (HC3) standard errors, using the imputed predictor data (n = 54). Variance inflation factors were all below 1.2, so multicollinearity is not a concern. The model explains little of the variation (R^2 = 0.13, adjusted R^2 = 0.04; overall F-test p = 0.40). Study consistency has the largest effect, about 4.3 points per standard deviation, but its interval includes zero (p = 0.075); the other predictors have effects close to zero. This agrees with the correlation results: habits are only modestly linked to self-rated academic functioning in this sample.

Table 7. Regression of academic score on standardized predictors

Predictor (standardized)	Effect on academic score (points per SD)	95% CI	p
Physical activity	2.62	-1.14 to 6.38	0.172
Sleep hours	-1.10	-4.82 to 2.62	0.563
Screen time	-0.26	-3.72 to 3.21	0.885
Study hours	-0.78	-6.21 to 4.66	0.779
Study consistency	4.26	-0.44 to 8.95	0.075

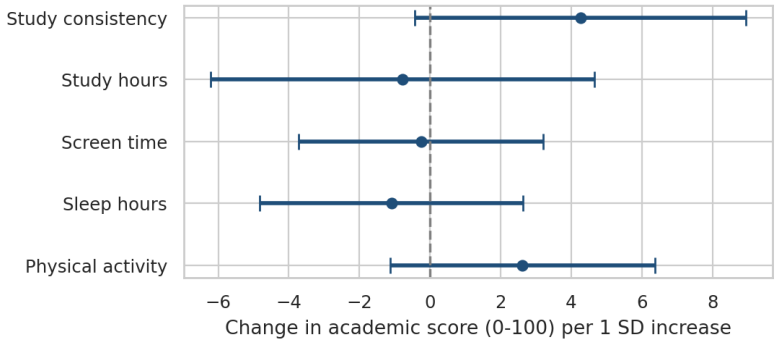


Figure 8. Regression coefficients with 95% confidence intervals (HC3 robust). Intervals crossing the dashed zero line indicate effects not distinguishable from zero.

5. Conclusion

5.1 Summary of findings

In this sample of 54 students, concentration was the outcome most clearly connected to everyday habits. Students with more consistent study routines (rho = 0.45) and better sleep quality (rho = 0.41) reported higher concentration. Sleep *duration*, physical activity and recreational screen time showed no statistically significant association with stress, productivity or CGPA, although more active students tended to report higher concentration and lower stress. Hypothesis by hypothesis: H1 was not supported; H2 was supported only for sleep quality; H3 was supported for phone distraction (students with up to 3 hours of recreational screen time reported markedly less) but not for productivity or concentration; and H4 was supported for concentration, not for CGPA. CGPA was unrelated to every lifestyle measure, and a regression of the academic score on all habits explained little (adjusted R^2 = 0.04). Descriptively, the typical student sleeps about 6.2 hours, spends about 6 hours a day on recreational screens, rarely follows a consistent study routine and reports high tiredness and stress.

5.2 Practical implications

- Improving **how well** students sleep (for example, a regular bedtime and less phone use at night) may matter more for focus than simply sleeping longer.
- A **regular daily study routine** is the habit most strongly linked to concentration, and is a realistic target for study-skills programmes.
- Students believe exercise helps their studies, but this sample neither confirms nor rules it out; wellness programmes should not promise academic gains from exercise alone.

5.3 Limitations

- **Small sample.** With n = 54, only correlations of about 0.37 or larger are reliably detectable; null results mean 'no evidence found', not 'no effect'.
- **Convenience sample.** Respondents were self-selected, 89% were male and 59% were in third year; the findings cannot be generalized to all students.
- **Self-reported data.** Hours, ratings and CGPA depend on recall and honesty; ratings are subjective and the wellbeing items were not a reliable single scale.
- **Correlation is not causation.** Students who feel focused may exercise or sleep better, rather than the reverse.
- **Data quality.** Numeric questions had no input validation, so answers arrived as free text and some values were unusable; many tests were run, so adjusted p-values should be preferred over raw ones.

5.4 Future work

A larger and more balanced sample, numeric input validation in the form, validated stress and sleep scales, and objective measures such as phone usage logs or wearable sleep data would allow stronger conclusions, including tests of whether the observed links hold within gender and year groups.

Appendix A. Final Questionnaire

Student Lifestyle and Academic Productivity Survey (Google Form, anonymous, 21 questions). Rating questions use a 1 to 5 scale. Question wording is taken from the form export; the answer options listed are those that appear in the collected responses.

Block	Question	Answer format
Profile	1. What is your current year of study?	1st year / 2nd year / 3rd year / 4th year / Postgraduate
	2. What is your gender?	Male / Female / Prefer not to say
Physical activity	3. On average, how many days per week do you go to the gym or do structured workouts (weights, cardio, yoga, home workouts, walking or jogging)?	0 days / 1-2 days / 3-4 days / 5 or more days
	4. On average, how many hours per week do you play sports (cricket, football, badminton, etc.)?	0 hours / 1-3 hours / 4-6 hours / 7 or more hours
Sleep and rest	5. On average, how many hours do you sleep per night on weekdays?	Short answer (number)
	6. At what time do you usually fall asleep on weekdays?	Before 11 PM / 11 PM - 1 AM / 1 AM - 3 AM / After 3 AM
	7. How would you rate your sleep quality over the past month?	Scale 1-5
	8. Do you use your phone in the last 30 minutes before sleeping?	Never / Sometimes / Always
Digital behavior	9. On average, how many hours per day do you spend on your phone/laptop for recreation (social media, videos, OTT, gaming, browsing)?	Short answer (hours)
	10. Out of that, how many hours per day do you spend on social media (Instagram, YouTube Shorts, Snapchat, etc.)?	Short answer (hours)
	11. How many hours per day do you spend on gaming or OTT (Netflix, Prime, etc.)?	Short answer (hours)
	12. How often do you get distracted by your phone while studying?	Scale 1-5
Study habits	13. On average, how many hours per day do you spend studying outside class (including assignments and projects)?	Short answer (hours)
	14. How consistent are you in your study routine?	Scale 1-5
	15. What is your approximate class attendance?	Below 60% / 60-75% / 76-90% / Above 90%
	16. What is your latest SGPA/CGPA (out of 10)?	Short answer (number)
Well-being	17. How would you rate your stress level during the last 4 weeks?	Scale 1-5
	18. How well are you able to concentrate while studying?	Scale 1-5
	19. How productive do you feel in your academic work (completing tasks on time, understanding concepts)?	Scale 1-5
	20. How often do you feel mentally tired or drained during the day?	Scale 1-5

Block	Question	Answer format
	21. Do you feel that your physical activity (gym/sports) helps your studies?	Yes, clearly / Somewhat / No effect / I don't do any physical activity